

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO2: Construct top-down and bottom up parsers using CFG for parsing conflicts and error recovery for efficient syntax tree generation. (BL3)

Assessment Tool Weightage: 10%

Duration :60 Mins

QUESTIONS: 3

Total Marks:50

Annexure A

SCENARIO BASED TASK

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

SCENARIO BASED TASK

S.No	Questions
1	A compiler developer is designing a parser for a programming language that supports nested arithmetic expressions using parentheses and operators. The developer must ensure that the grammar correctly generates valid expressions such as: (id + id) * id id + (id * (id + id)) However, during implementation, the developer realizes that the grammar must be carefully designed to: • Maintain correct operator precedence • Support nested structures • Avoid ambiguity In this scenario, the developer must design and analyze a suitable Context-Free Grammar (CFG) for the language. Tasks: (a) Construct a Context-Free Grammar (CFG) that generates valid arithmetic expressions using: • Identifiers (id) • Operators (+, *) • Parentheses (,) (b) Explain how your grammar ensures: • Correct operator precedence (* > +) • Proper handling of nested expressions (c) Determine whether the grammar is ambiguous or unambiguous. Justify your answer with an example. (d) Show the derivation (leftmost or rightmost) for the input string: id + id * id (e) Construct the parse tree for the above expression. (f) Suggest how the grammar can be modified if ambiguity exists.

A compiler developer is designing a shift-reduce parser for a language that generates balanced patterns of 0's and 1's. The language ensures that every string starts and ends with matching symbols in a structured manner.

The developer uses the following grammar:

$$S \rightarrow 0 S 1 \mid 01$$

The parser must analyze whether a given input string belongs to the language by applying bottom-up parsing techniques.

The input string is:

0011

Tasks:

(a) Explain the concept of shift-reduce parsing and how it works in bottom-up parsing.

(b) Identify the handles in the input string during the parsing process.

(c) Perform step-by-step shift-reduce parsing for the input string:

- Show stack contents
- Show remaining input
- Indicate actions (shift / reduce / accept)

(d) Construct the rightmost derivation in reverse for the input string.

(e) Determine whether the given grammar is ambiguous or unambiguous, and justify your answer.

(f) Suggest how the parser detects errors if the input string is invalid (e.g., 0101).

RUBRICS FOR CALCULATION:

Scenario based assessment

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand